# Khulna University of Engineering & Technology

Project on

## AI-Driven Glucose Monitoring System for Telemedicine Application

**Course no: BME 4232**

**Course title: Telemedicine and Health Care Laboratory**

**Supervised by:**


**Siddika Tamanna Islam**
Lecturer
Department of Biomedical Engineering
Khulna University of Engineering & Technology

**S. M. Amanul Haque**
Lecturer
Department of Biomedical Engineering
Khulna University of Engineering & Technology

# Group members




**Kazi Nimat Hasan**
**2015007**



**Shaira Afra Redhila**
**2015008**



**S.M. Rafiur Rahman**
**2015009**



**Sraboni Hossain**
**2015010**



**Labonno Rahman**
**2015011**



**Prionty Majumder**
**2015012**

# Motivation (1/1)

Diabetes is a major global health problem and requires frequent blood glucose monitoring.

Conventional glucose monitoring methods:

- Invasive (painful, infection risk)
- Expensive (especially CGMs)
- Limited accessibilty in rural/low-resource settings (like rural areas)

Advantages of our approach:

- Low-cost, inexpensive spectrophotometer
- Simple and accurate analysis
- Standardized method
- AI integration



**Figure 1.1. Conventional blood glucose monitoring**

# Contribution (1/1)

| Group member | Roll | Contribution |
|---|---|---|
| Kazi Nimat Hasan | 2015007 | Circuit setup, Sample Solution Preparation, Data collection, Dataset formation, Report writing & Presentation slide preparation |
| Shaira Afra Redhila | 2015008 | Circuit setup, Sample Solution Preparation, Data collection, Dataset formation & Presentation slide preparation |
| S. M. Rafiur Rahman | 2015009 | Circuit setup, Sample Solution Preparation, Data collection, Dataset formation, Report writing & Presentation slide preparation |
| Sraboni Hossain | 2015010 | AI model, app development & Presentation slide preparation |
| Labonno Rahman | 2015011 | AI model, app development Presentation slide preparation |
| Prionty Majumder | 2015012 | Sample Solution Preparation, Report writing & Presentation slide preparation |

# Contents


01 Objectives

02 Introduction

03 Methodology

04 Application Development

05 Conclusion

# Objectives (1/1)

The primary objectives of our project are as follows:

- ❑ To design and develop a minimally invasive glucose estimation system using spectrophotometric techniques.
- ❑ To collect and pre-process the data from the setup and train a machine learning model for trustworthy prediction of glucose levels.
- ❑ To develop a telemedicine application for estimating glucose levels and make it accessible for healthcare professionals and other caregivers.

# Introduction (1/3)

## ❑ EXISTING glucose monitoring approaches

| | |
|---|---|
| Electrochemical Methods<br>(Most common today) | • Finger-prick glucometers<br>• Continuous Glucose Monitors (CGMs)<br>➢ **Limitation:**<br>Measures current, not direct optical properties and can be affected by temperature, pH, interfering substances |
| Optical Methods<br>(Research + Lab-based) | • Spectrophotometry<br>• Near-infrared (NIR) sensing<br>• Raman spectroscopy |

## ❑ Is spectrophotometric glucose monitoring actually used?

- Mainly in laboratories, not handheld devices.
- Clinical biochemistry labs
- Automated analyzers

# Introduction (2/3)

## ❑ GOD-POD Method for Glucose Detection

✓ Enzymatic colorimetric method for glucose estimation

$$\text{Glucose} + O_2 + \text{GOD} \rightarrow \text{Gluconic acid} + H_2O_2$$

$$H_2O_2 + \text{POD} \rightarrow \text{Colored solution}$$

✓ Converts glucose into a measurable color signal

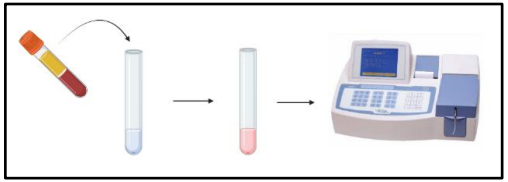


**Figure 1.2: Estimation of Blood Glucose by GOD-POD method**

## ❑ Spectrophotometric Measurement Principle (Based on Beer-Lambert Law)

- ✓ Light passes through sample
- ✓ Some light is absorbed and remaining light is detected
- ✓ Phototransistor measures transmitted light
- ✓ Higher absorbance → Higher glucose concentration
- ✓ Colour intensity is converted into quantitative glucose value
- ✓ More glucose → more color intensity
- ✓ Color ∝ Glucose concentration

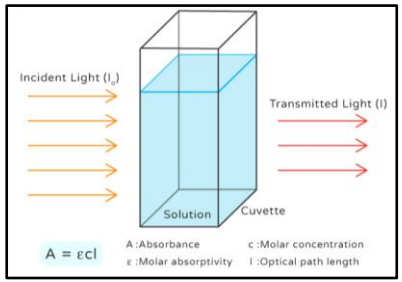



**Figure 1.3: Measurement Principle**

# Introduction (3/3)

## ❑ Machine Learning in Healthcare

- ✓ Enables data-driven medical prediction
- ✓ Enhances accuracy and analytical efficiency
- ✓ Supports interpretable healthcare decision-making

## ❑ Mobile Application & Telemedicine

- ✓ Facilitates remote healthcare monitoring
- ✓ Improves accessibility of medical services
- ✓ Enables real-time digital health management

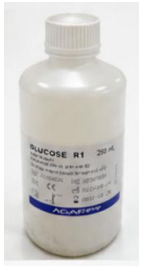


**Figure 1.4. Mobile Healthcare and Telemedicine Application**

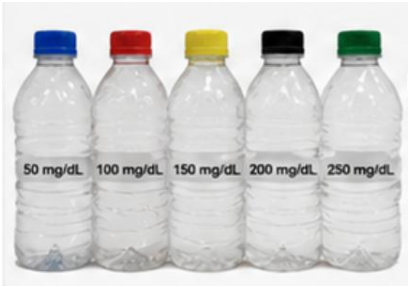
# Methodology (1/4)

## ❑ Hardware-based sensing

### 1. Sample Preparation



a) Reagent bottle

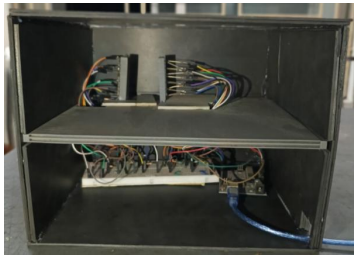



b) Glucose solutions of varying concentrations

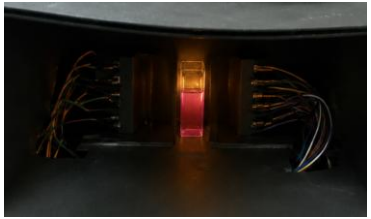


c) Glucose solutions mixed with GOD-POD reagent in a cuvette

### 2. Solution Testing

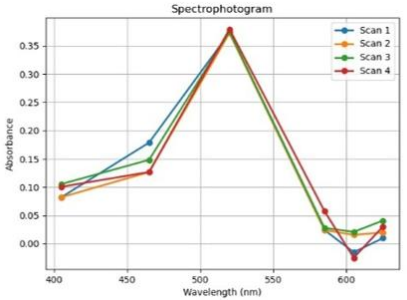


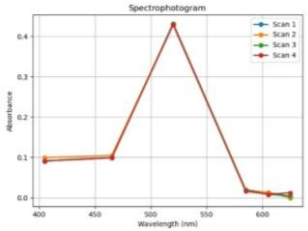
c) Hardware Setup of the Proposed System



d) Spectrophotometric scanning (inside dark compartment)

### 3. Absorbance spectrum

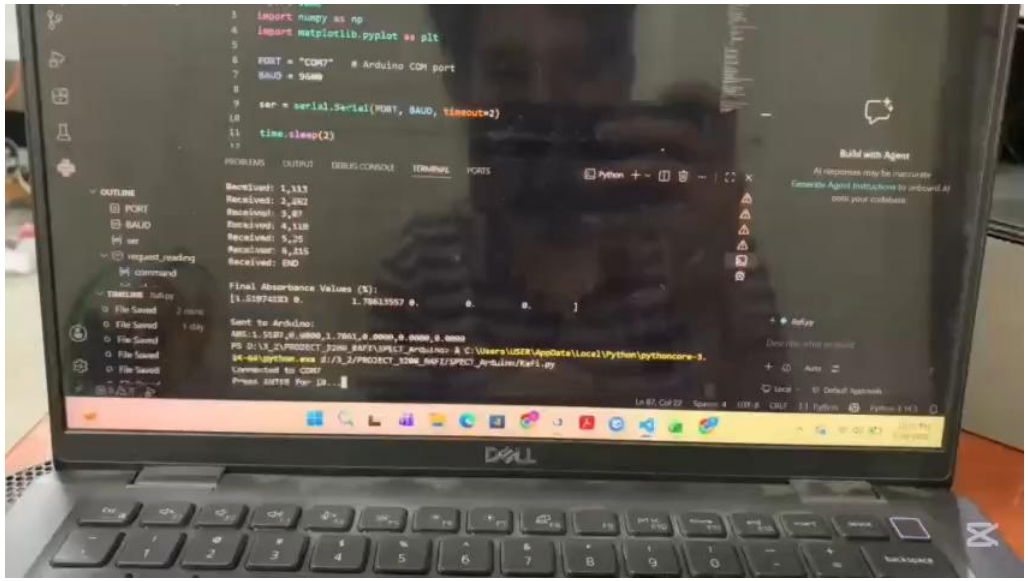



e) Spectrophotogram of varying concentrations

# Methodology (2/4)

- **Hardware-based sensing**




**Video 1: Working Process of the Project**

# Methodology (3/4)

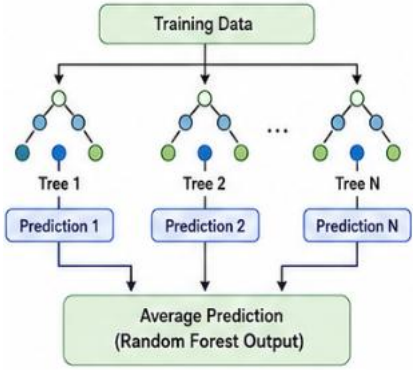
## ❑ Machine Learning-Based Prediction

### 3. Dataset generation

| Glucose Concentration (per 100 ml) | Wavelength Features (nm) | | | | | |
|---|---|---|---|---|---|---|
| | 420 | 460 | 520 | 570 | 610 | 630 |
| 50 | 1.12 | 3.70 | 12.78 | 0.51 | 1.19 | 1.24 |
| 50 | 3.82 | 5.67 | 12.78 | 0.00 | 1.21 | 1.27 |
| 50 | 1.87 | 4.67 | 12.90 | 1.14 | 1.28 | 1.16 |
| 50 | 0.00 | 0.90 | 12.67 | 1.18 | 1.25 | 1.11 |
| ... | ... | ... | ... | ... | ... | ... |

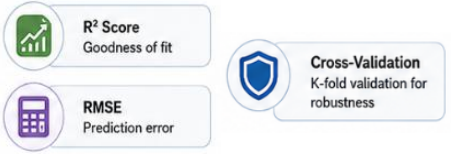
f) Absorbance readings at different wavelengths for varying concentrations

### 4. Data Analysis

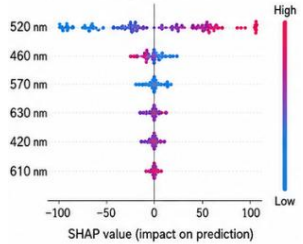



g) Random Forest Regression

### 5. Model evaluation




h) Assessment of model performance




i) SHAP analysis

# Methodology (4/4)

## ❑ Mobile Application Integration

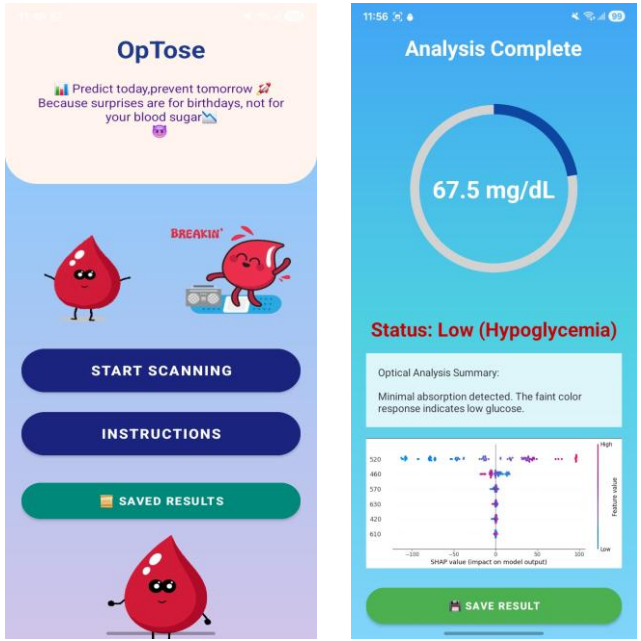
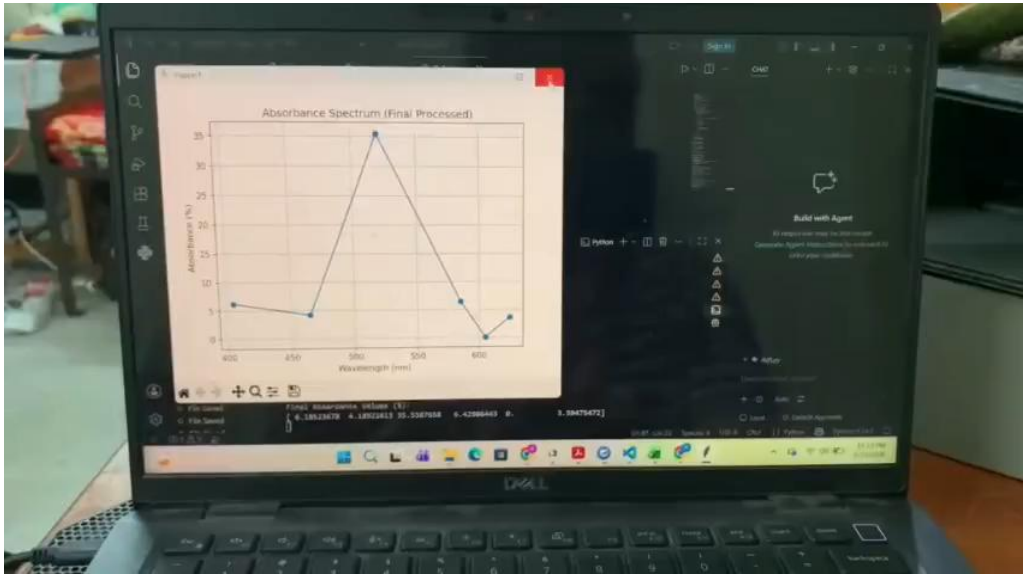



Figure 2.2: Application Connectivity with the Project




Video 2: Demonstration of the Application Connectivity

# Hardware Setup (1/3)

## Circuit & Enclosure

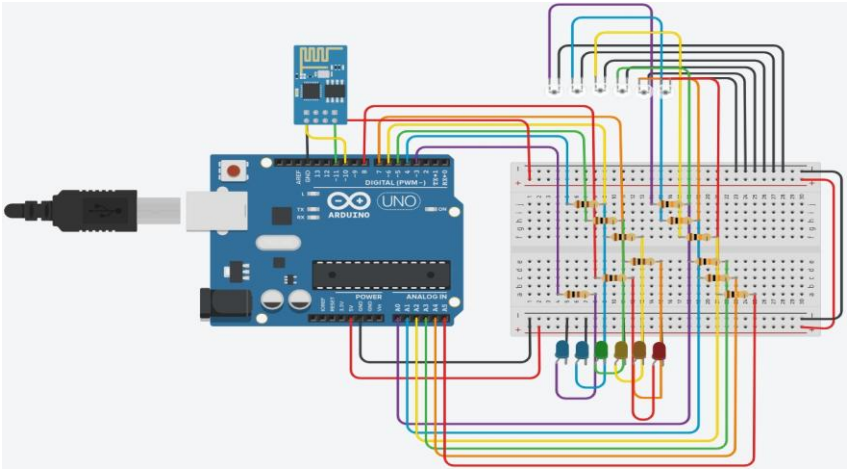



Figure 3.1: Circuit Implementation in TinkerCAD

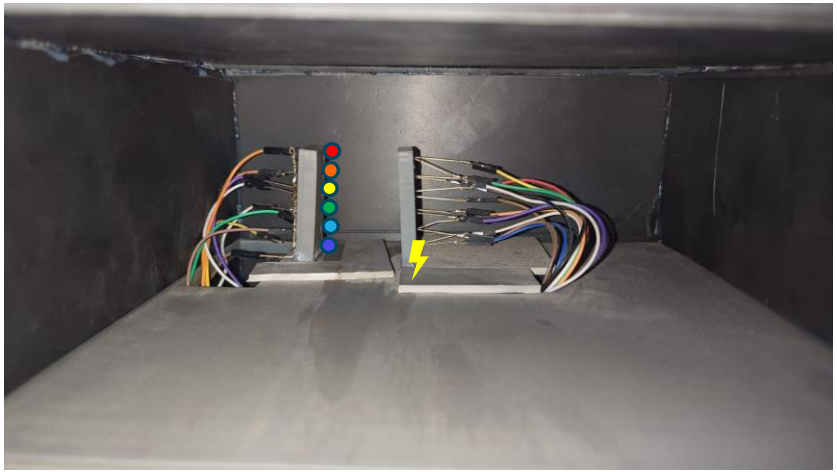


Figure 3.2: Physical arrangement of the emitter-detector sensing chamber

# Hardware Setup (2/3)

## Dataset Generation

- Each time 4 scans were being made
- Each concentration solution was being tested 6 times
- Total of 24 data for each concentration
- In total of 200 rows of data were being there finally

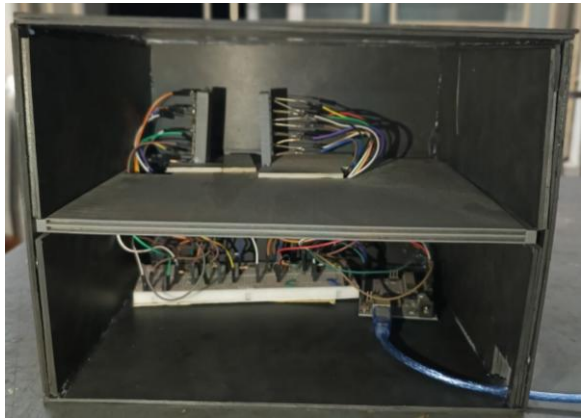


**Figure 3.3: Double Compartment Contained Enclosure**

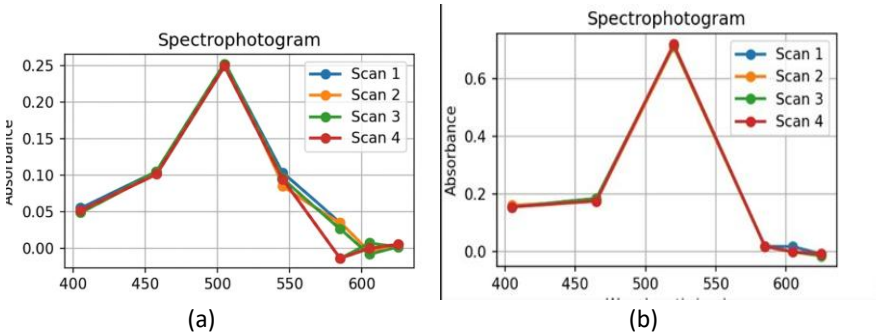



**Figure : Gained absorption data for glucose concentration of**

| entration(per 1 | Readings | 420 | 460 | 520 | 570 | 610 | 630 |
|---|---|---|---|---|---|---|---|
| 50mg | | 1.12 | 3.7 | 12.78 | 0.51 | 1.19 | 1.24 |
| | | 3.82 | 5.67 | 12.78 | 0 | 1.21 | 1.27 |
| | | 1.87 | 4.67 | 12.9 | 1.14 | 1.28 | 1.16 |
| | 1st | 0 | 0.9 | 12.67 | 1.18 | 1.25 | 1.11 |
| | | 4.58 | 12.49 | 18.14 | 0 | 0 | 0 |
| | | 4.14 | 12.49 | 18.28 | 1.1 | 1.16 | 1.22 |
| | | 3.28 | 11.54 | 17.87 | 1.1 | 0 | 1.18 |
| | 2nd | 3.28 | 10.61 | 17.87 | 0 | 0 | 1.24 |
| 220 mg | | 1.15 | 3.1 | 45.37 | 1.22 | 1.28 | 1.19 |
| | | 0.4 | 4.19 | 46.43 | 1.18 | 1.24 | 1.19 |
| | | 0.4 | 5.87 | 45.11 | 1.22 | 1.24 | 1.26 |
| | 1st | 1.14 | 4.19 | 44.33 | 1.2 | 1.27 | 1.26 |

**Figure 3.4:Partial portion of initial dataset**

# Hardware Setup (3/3)

## Wireless Transmission

- Absorbance measured at six wavelengths
- Data sent via Bluetooth (HC-05) module
- Data Format: A1, A2, A3, A4, A5, A6
- Real-time data transfer to mobile app

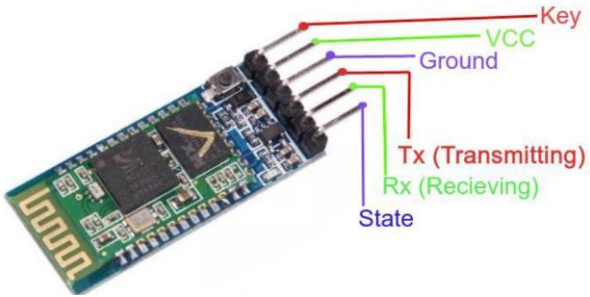



**Figure 3.5: Representation of HC 05 Module**

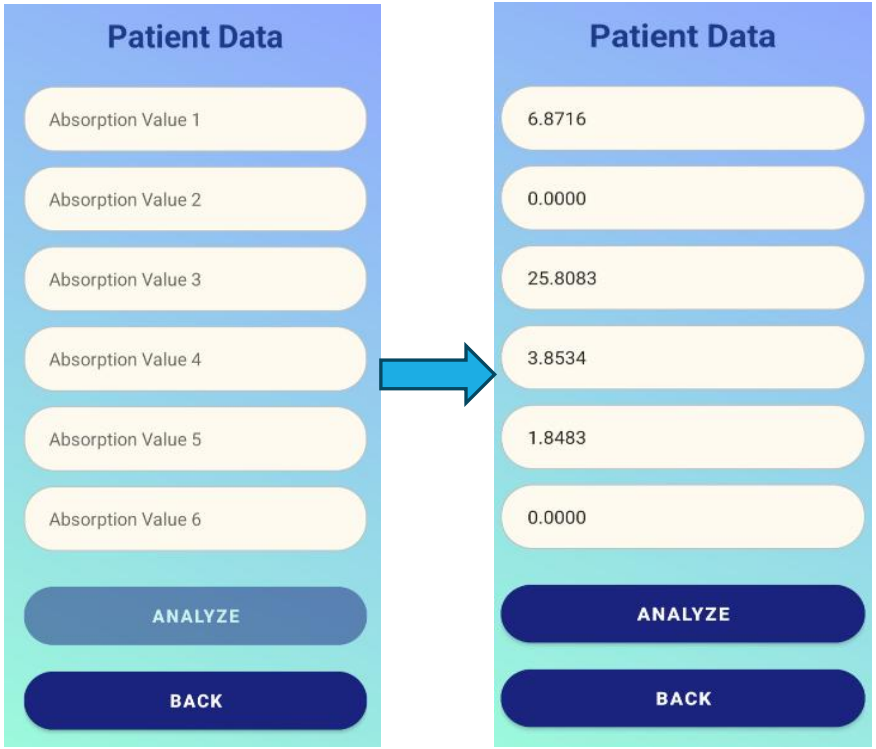



**Figure 3.6: Representation of data transfer validating interface**

# Dataset Preprocessing (1/1)

## ❑ Dataset Augmentation

- Limited dataset size could cause overfitting and poor generalization.
- MixUp augmentation was used to generate synthetic samples.
- Two random samples were combined using weighted interpolation.
- 220 synthetic samples were generated to improve dataset robustness.

**MixUp Formula :** $x_{new} = \lambda.x_i + (1-\lambda)\, x_j$

| Concentration(per 100 ml) | 420 | 460 | 520 | 570 | 610 | 630 |
|---|---|---|---|---|---|---|
| 147.0913792 | 11.9302 | 16.8756 | 48.2908 | 0.5768 | 1.22352 | 1.26831 |
| 250 | 15.8736 | 17.6095 | 71.0242 | 1.95619 | 0.7641 | 1.1859 |
| 220 | 6.4315 | 10.8564 | 48.5806 | 1.19471 | 0.99375 | 1.14229 |
| 101.5036759 | 10.4335 | 13.7026 | 29.6711 | 2.19655 | 4.32371 | 0.64341 |
| 150 | 5.14192 | 15.0344 | 36.783 | 1.94121 | 1.24149 | 1.07844 |
| 220 | 0.4 | 4.19 | 46.43 | 1.18 | 1.24 | 1.19 |
| 238.3241098 | 17.56 | 25.8897 | 65.3179 | 2.20378 | 1.27926 | 1.12899 |
| 173.1207768 | 3.94261 | 7.90983 | 40.9513 | 1.16651 | 1.25383 | 1.24262 |
| 110 | 6.45 | 9.14 | 30.61 | 0.67 | 1.1 | 0 |
| 220 | 1.15 | 3.1 | 45.37 | 1.22 | 1.28 | 1.19 |

**Figure 4.1: Preprocessed Augmented Dataset**

## ❑ Gaussian Noise Addition

- Gaussian noise was added to simulate real sensor variability.
- Mean = 0 and standard deviation = 0.01.
- Final dataset was shuffled before model training to remove ordering bias.

**Gaussian Noise Distribution :** $X_{noisy}$ = X + N(0, $\sigma^2$)

# Machine Learning Model (1/1)

## ❑ Random Forest Regression

- Random Forest Regressor was used for glucose concentration prediction.
- Dataset was divided into : 80% training data
  20% testing data
- The model used multiple decision trees for ensemble learning.
- Captured non-linear relationship between glucose concentration and wavelengths.
- Final prediction was obtained by averaging outputs from all trees.

## ❑ Model Optimization

- Hyperparameters were manually tuned for improved performance.
- n_estimators = 150 was selected for stable prediction.
- max_depth = 8 was used to reduce overfitting.

**Prediction Formula : y** $= \frac{1}{N} \sum_{i=1}^{N} T_i\,(x)$

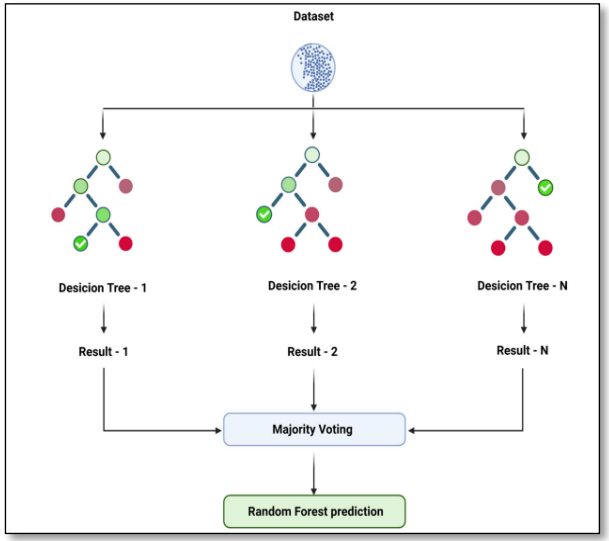



**Figure 4.2: Schematic representation of Random Forest model**

# Model Evaluation & Prediction (1/1)

## ❑ Performance Evaluation

- Model performance was evaluated using $R^2$ score and root mean square error value (RMSE).
- 5-fold cross-validation was applied to ensure reliability and stability.

```
Test R2: 0.960
RMSE: 12.511
```

**Figure 4.3: Model Evaluation Results**

```
CV R2 scores: ['0.977', '0.953', '0.951', '0.934', '0.825']
Mean CV R2: 0.928
```

**Figure 4.4: Cross-Validation Results**

## ❑ Glucose Classification

- Predicted glucose values were converted into clinical categories:
  - -Hypoglycemic (< 71 mg/dL)
  - -Normal (71-100 mg/dL)
  - -Pre-diabetic (101-125 mg/dL)
  - -Diabetic (>125mg/dL)

```
Predicted Glucose: 110.2
Condition: Pre-Diabetic
```

**Figure 4.5: Glucose Prediction & Classification**

# Model Interpretability (1/1)

## ❑ Feature Importance Analysis Using SHAP

- SHapley Additive exPlanations analysis identified the contribution of each wavelength feature to glucose prediction.
- The 520 nm feature showed the highest impact on model output.
- SHAP improved interpretability of the Random Forest model.

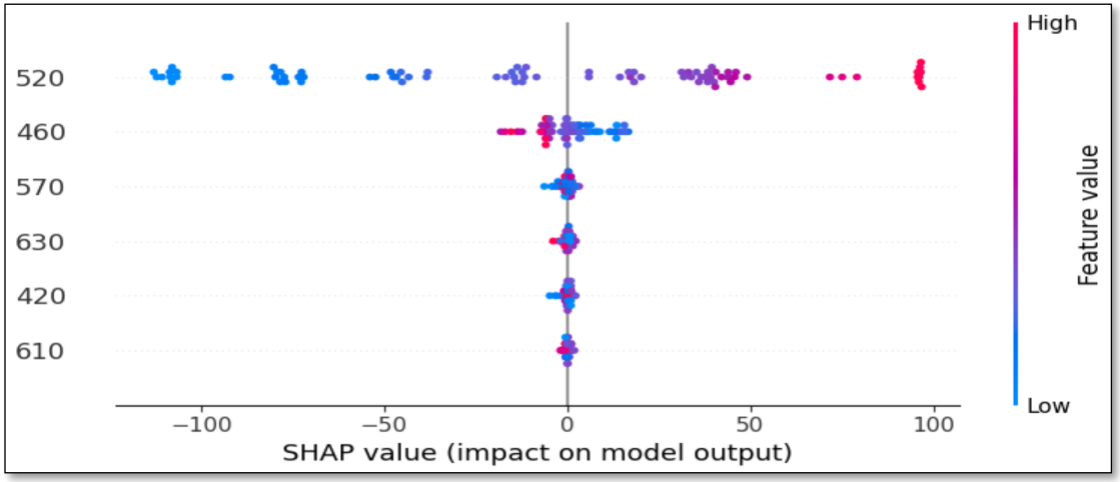



**Figure 4.6: SHAP Summary Plot of Wavelength Features**

# Application Development(1/4)

**What is OpTose?**

- An android application developed for monitoring blood glucose level.
- A Mobile-First IoT Integration approach has been utilized.
- Kotlin and XML (Extensible Markup Language ) has been used for logic and UI design respectively.
- The AI model is converted to Java before integration for its light-weight feature.

OpTose

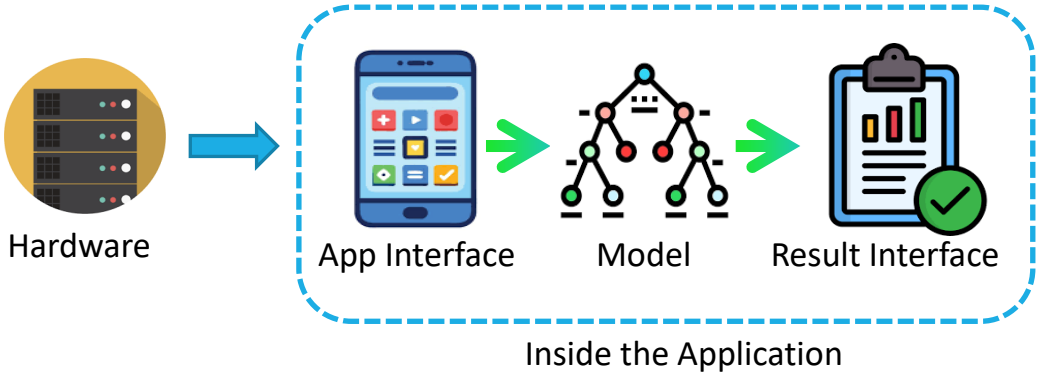



Figure 5.1: Data-flow from Hardware to Software

# Application Development(2/4)

## Home Interface

- Has three buttons that lead to three different pages.
- Text box transition made by Translate Animation tool.

## Instruction Interface

- Provides basic instructions on how to use the app and AI interpretation.
- Also includes some other basic information.

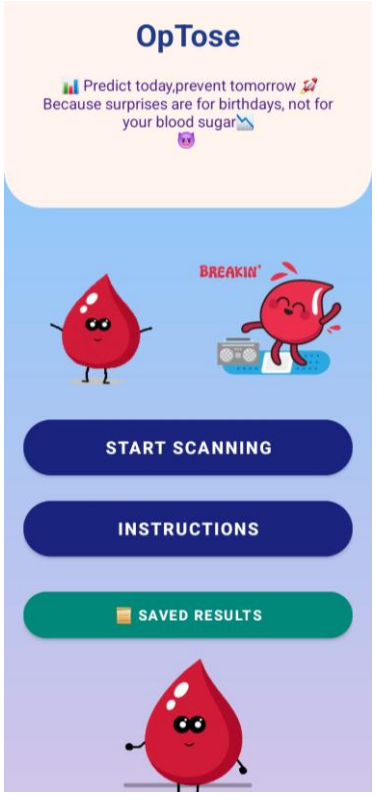



Figure 5.2: Home Page

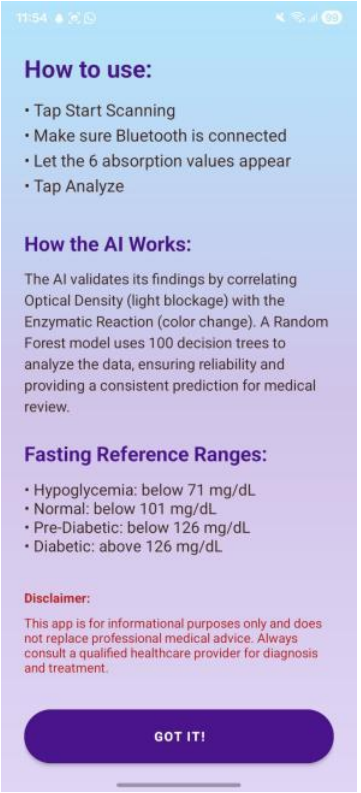



Figure 5.3: Instruction Page

# Application Development(3/4)

## Save Result Interface

- Contains patient information (name) and data (result) with date.
- Records can be sent to the patient through email, deleted manually or all at once for convenience.

## Input Interface

- Transition from left to right of the text boxes made by using staggered postDelayed animations tool.
- Asks for the Bluetooth to be turned on, when it's on, six absorption values appear atomatically one by one and the last one triggers the analyze button, then it pops up for analysis.




Figure 5.4: Input Page

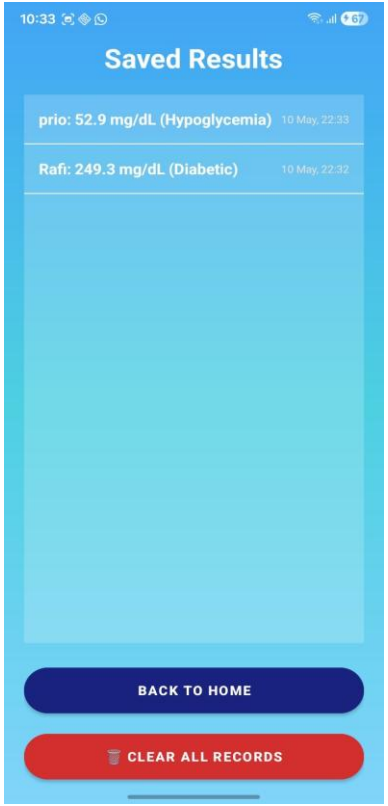



Figure 5.5: Saved Results Page

# Application Development(4/4)

## Result Interface

- Represents the glucose concentration level, status, optical analysis and SHAP plot.
- Animated Gauge Visualization is made using circular progress bar with a custom progress drawable.
- Has three buttons for saving the results, sending the results to the patient through email and going back to home page.




Figure 5.6: Result Page Buttons

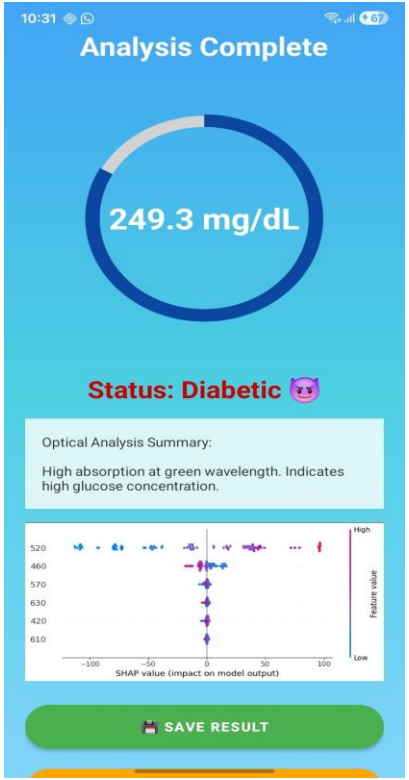



Figure 5.7: Result Page

# Conclusion (1/3)

## A Quick Overview & Advantages



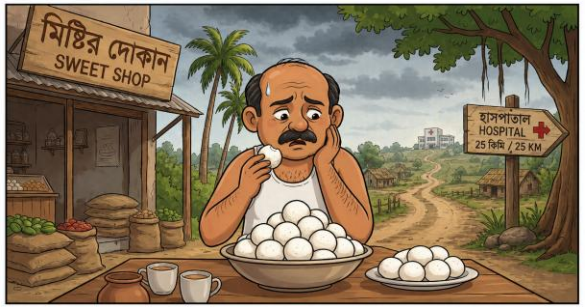




Figure 6.1 :Story of Rahim uncle

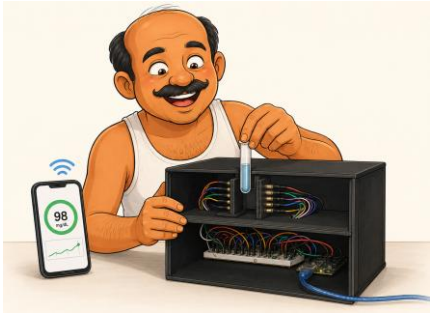






Figure 6.2:Remote Monitoring




Figure 6.3: No need of Glucose strips



Figure 6.4: Cost Effective

# Conclusion (2/3)

## Limitations & Future Works



Figure 6.4: Continuous source of light

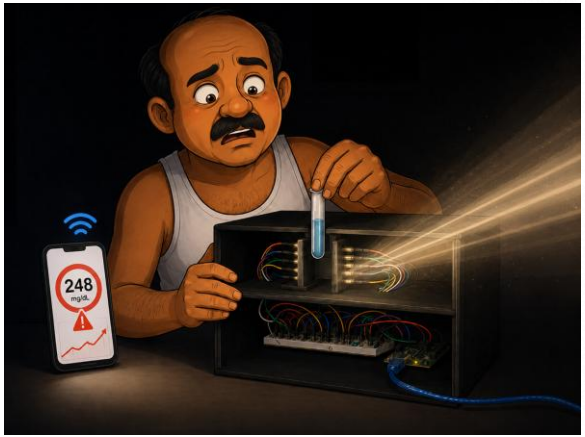



Figure 6.5: Needs Dark Environment



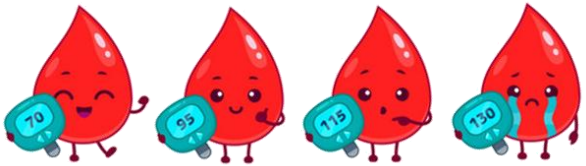
Figure 6.6: Small dataset



Figure 6.7:Cheap Sensors

# Conclusion (3/3)

"Predict today, Prevent Tomorrow.
Because surprises are for Birthdays,
not for your blood sugar."



**Thank You for your patience**
**Feel free to ask any question**